

Safety and Reliability Training

Module 2: Reliability Engineering

Submodule 2.A : Reliability Prediction

Reliability Prediction





Objectives of this training

✓ To know some Reliability Prediction principles





Outline

- ☐ Introduction
- ☐ Purpose
- ☐ Definitions
- ☐ Failure Rate vs Time
- ☐ Useful Life
- ☐ Reliability
- ☐ Reliability Prediction
- ☐ Reliability Modelling - Introduction
- ☐ Reliability Modelling – System Definition



Outline

- ☐ Reliability Modelling – Operational Conditions
- ☐ Reliability Modelling – Model Construction
- ☐ Reliability Modelling – Basic Models
- ☐ Reliability Prediction – Methods
- ☐ Reliability Prediction – Similar Equipment
- ☐ Reliability Prediction - Part Count Method
- ☐ Reliability Prediction - Part Stress Method
- ☐ Reliability Prediction – MIL-HDBK-217F Part Model
- ☐ Reliability Prediction – FIDES Part Model
- ☐ Reliability Prediction – Telcordia Part Model

Introduction

- Reliability prediction is a quantitative analysis technique used to predict the Reliability behavior of a system under specified operating conditions.
- It is a continuing activity throughout a project, with the prediction being regularly updated as more design, test and evaluation data become available. The accuracy of any prediction depends largely upon the availability of detailed design and operating data.
- Clearly, therefore, prediction should be a part of the design process and not simply a parallel or even a later activity.

Purpose

- Evaluate whether or not a particular design concept is likely to meet a specified reliability requirement under defined conditions.
- Assist in the identification and elimination of any potential reliability problems by imposing a systematic discipline that ensures all reliability aspects of a design are examined.
- Compare alternative design solutions.
- Provide inputs to related project activities, such as:
 - Design evaluation.
 - Trade-off studies.
 - Safety analyses.
 - Spares provisioning.
 - Logistic and maintenance support studies.

Definitions

- **Reliability**

- (Concept) The ability of an item to perform a required function without failure under stated conditions for a stated period of time.
- (Quantitative measure) The probability that an item can perform a required function under given conditions for a given time interval, $(t1, t2)$. This is normally denoted either by the letter R or by $R(t)$, with t denoting the interval $(t1, t2)$.



Definitions

- **Failure**

- The state of the item when it is unable to perform a required function. In the case of non-repairable items, it is the termination of the ability of an item to perform a required function.

Notes:

- 1. After the occurrence of a failure, the item is in a faulty condition.
- 2. An occurrence of a failure is an event (as distinguished from a fault, which is a state).
- 3. This concept as defined does not apply to items consisting of software only.



Definitions

- **(Instantaneous) Failure Rate**

- The limit, if exists, of the ratio of the conditional probability that the instant of time, t , of a failure of an item falls within a given time interval, to the length of this interval, $(t, t + \Delta t)$, when tends to zero, given that the item is in an up state at the beginning of the time interval.
- This limit is normally denoted by $\lambda(t)$. Failure rates are often given in terms of failures per million hours (fpmh); however, some industries use an alternative measure of failures per known as FITs (Failures in Time, failures per billion hours).

Definitions

- **Mean Time Between Failures (MTBF)**

- MTBF is a statistical estimate of the time, usually measured in hours, between failures of an item or system. It is commonly provided by manufacturers, but MTBF comparison can be misleading since:
 - Each MTBF calculation standard has its own set of assumptions, which can significantly impact the outcome.
 - Databases are not accurate.
 - Application conditions can be very different.
- When dealing with non-reparable items, Mean Time To Failure (MTTF) metric is used.

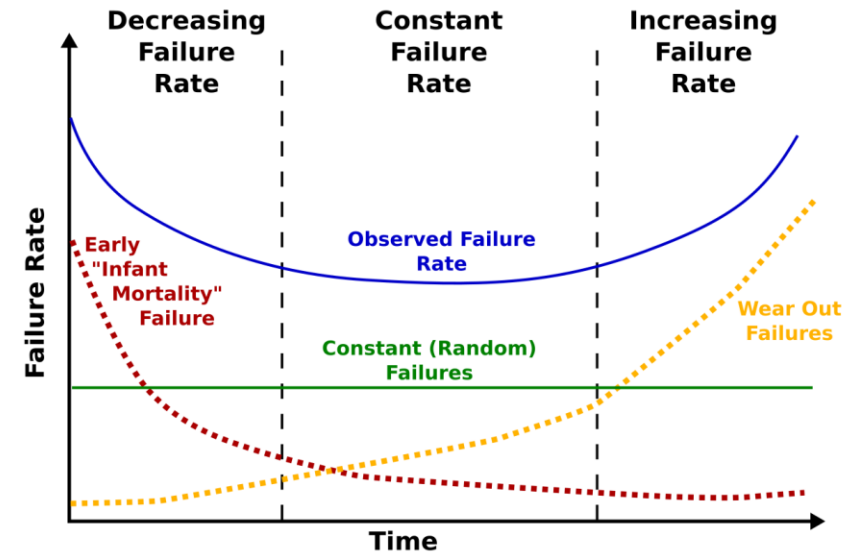
Failure Rate vs Time

- A generally accepted and simplified model for variation of failure rate with time is the classical bathtub curve.
- However, it is difficult to know where a product is along the bathtub curve, or even if the bathtub curve is applicable to a certain product without large amounts of products in use and associated failure rate data.



Failure Rate vs Time

- Initially, failures of an item placed in service are often seen to be dominated by 'quality' failures. These failures occur in what is known as the 'infant mortality' period.
- Later, as the system 'ages', the item enters a 'wear-out' period.
- The 'useful life' period, where there is often seen to be a 'constant' failure rate, lies between the infant mortality and wear-out periods.



Useful Life

- The useful period is the interval of most interest from a reliability prediction standpoint because, if a rigorous reliability programme is applied throughout a project lifetime, it is assumed that:
 - ✓ The majority of early life failures will normally be eliminated before an item enters service.
 - ✓ An in-service maintenance policy will ensure that items are replaced before wear-out becomes a significant problem.
- A prediction based on the useful life, in general, represent the reliability of a 'mature' design whose failure rate comprises mainly stress-related failures. Where the assumptions above are not given proper consideration, predictions will be substantially optimistic.

Reliability

- Assuming that the conditions described in Useful Life Failure Period apply so that the failure rate is constant:

$$\mathbb{R}(t) = e^{-\lambda t}$$

$$MTBF = \frac{1}{\lambda}$$

Where:

- $R(t)$ is the probability that an item will survive for time under the specified operating conditions.
- λ is the item failure rate under the specified operating conditions of temperature, stress, environment, etc...
- t is the time that the item is at risk under the specified operation conditions. This is sometimes called the mission time.
- MTBF is the Mean Time Between Failures.

Reliability

- Providing that a system can perform its functions at some point in time, it will continue to have that capability until the operating characteristics of a component or part (or group of components/parts) changes to the extent that the specified function of the system is no longer achieved.
- The reliability of a system, therefore, depends upon:
 - The number of components and parts.
 - The way in which these components and parts are interconnected to perform the system functions.
 - The reliabilities of the individual components.



Reliability Prediction

- Reliability prediction is a quantitative analysis technique used to predict the failure rate of a system based on the components of the system and its operating conditions.
- It depends essentially upon producing a model that represents the reliability relationships between items comprising a system and then evaluating the various elements within the model to provide a quantitative estimate of the system reliability and its constituent parts.



Reliability Modelling - Introduction

- The purpose of reliability modelling is to express the specified requirements, functions, and operating and maintenance conditions for a system in such a way that the reliabilities of its items can be assessed and combined to predict the system reliability, indicate shortcomings and assess logistic implications.
- There can be as many reliability models for a particular system as failure definitions but the most useful ones are those which strike a good balance between an accurate representation of a real-life situation and the need to provide results in a reasonable time, with due regard to the quality and quantity of input data and the required accuracy of the results.

Reliability Modelling – System Definition

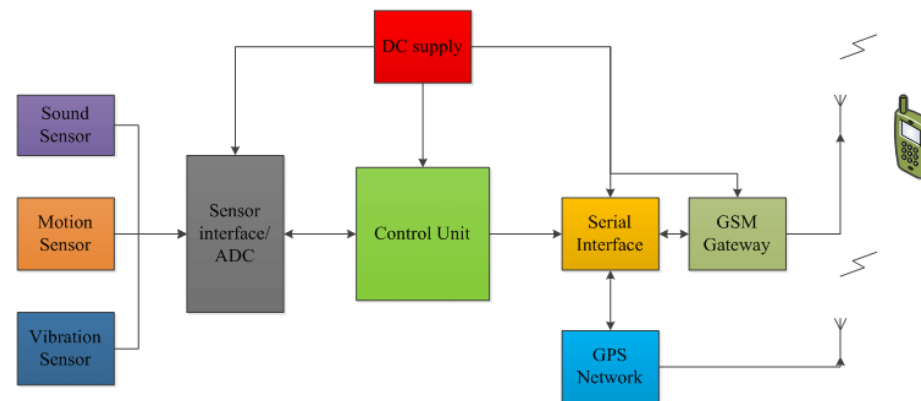
- The major sub-systems that comprise the system should be identified and their functional relationships established with respect to the system functions. If the functional configuration of the system can vary during operational use, then each configuration must be identified separately.
- The system performance requirements should be detailed and the conditions that constitute a system failure defined. If a particular failure condition applies only to a limited part of the operational duty cycle then this should be noted.



"The system's down."

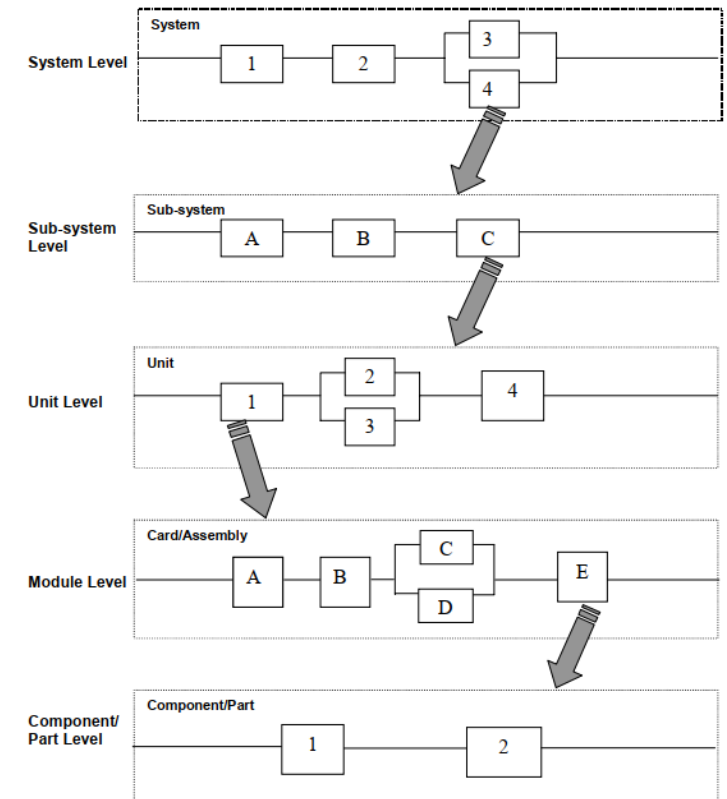
Reliability Modelling – System Definition

- For modelling purposes, the functional relationships within a system must be developed through successive levels of the assembly to the component/part level. For a large system, it is generally best to establish the relationships between the major sub-systems first, and then to consider each sub-system individually.
- Functional block diagrams (or other similar methods) should be used to show the functional relationships within a system in a concise and visual manner.



Reliability Modelling – System Definition

- In general terms, a system is a combination of items that are interconnected with each other to perform a specific operational function or functions.
- At its System highest level, there are several pieces of equipment, each designed to perform a particular function as a self-contained unit.
- Alternatively, at the lowest level of assembly, a system may be a combination of individual electronic or mechanical parts providing an input function to the next higher level of assembly.

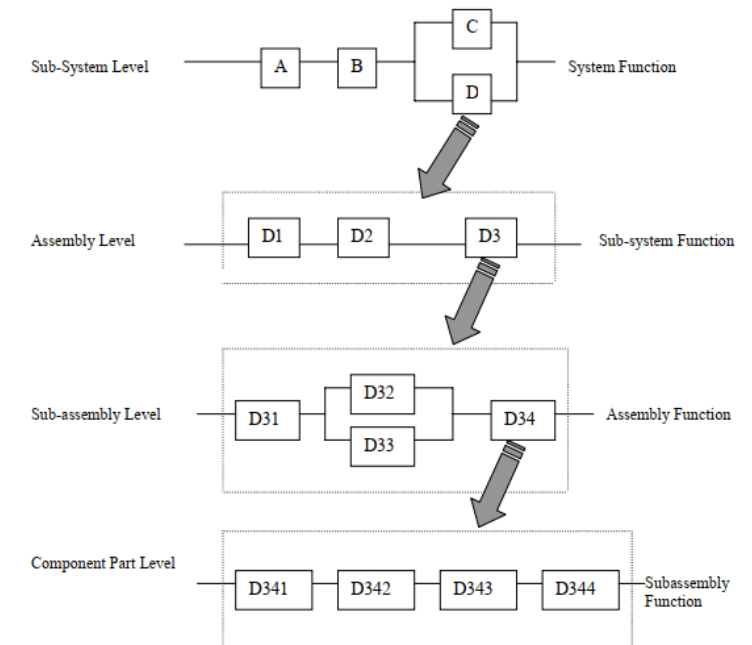


Reliability Modelling – Operational Conditions

- The purpose and functions of the system should be described. If a system has more than one functional mode of operation (e.g., an aircraft, a search and tracking radar, etc.), the requirements in each mode should be identified separately.
- The main performance, safety and physical characteristics should be listed in order of importance. Acceptable limits of satisfactory performance should be stated so that failure criteria can be established, and any acceptable performance degradation that still allows a limited operational capability should also be defined.

Reliability Modelling – Model Construction

- To construct a reliability model for a system, the reliability relationships between the items comprising the system must first be established. A common method of representing such reliability relationships within a system is a Reliability Block Diagram (RBD).
- Once an RBD has been constructed to show the reliability dependencies at the system level, a similar procedure should be followed to construct RBDs for each sub-system at successive levels of assembly down to the level at which reliabilities, or failure rates, can be estimated from the component/part data.



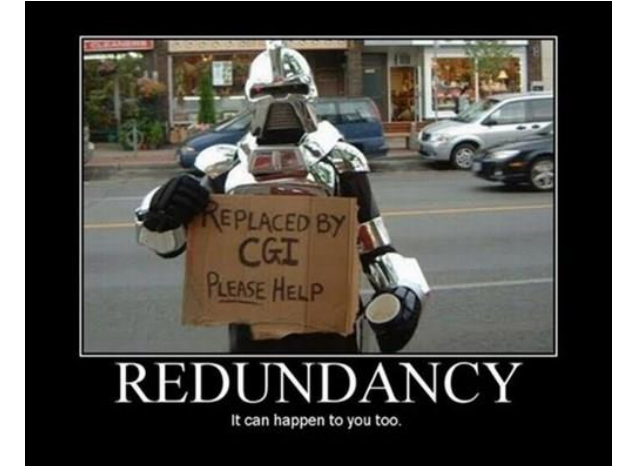
Reliability Modelling – Basic Models

- A reliability model for a group of items is derived by combining the reliabilities of the individual items according to set rules for combining probabilities. The two simplest configuration of items are:
 - **Series configuration.** All items must operate successfully for the group to be successful. Under the assumptions that each item failure is totally independent of another (this is to say, one item failure does not influence other's failure probability), and item failure rates are constant, total reliability is equal to the product of reliabilities and total failure rate is equal to the sum of failure rates:

$$R_s(t) = \prod_{i=1}^N R_i(t) = \prod_{i=1}^N e^{-\lambda_i t} \quad \lambda_s = \sum_{i=1}^N \lambda_i$$

Reliability Modelling – Basic Models

- **Redundant configuration.** There are three basic types:
 - Parallel Redundancy - Multiple component are always active. However, only one or more of these components are required to be active for system operation
 - Load-sharing Redundancy – The components all work together to handle the same load. When one component fails, the remaining operating components increase their amount of the load.
 - Standby Redundancy - Multiple components are available but not all of them are active. If the active component fails, a non-operational one is switched on to take its place.

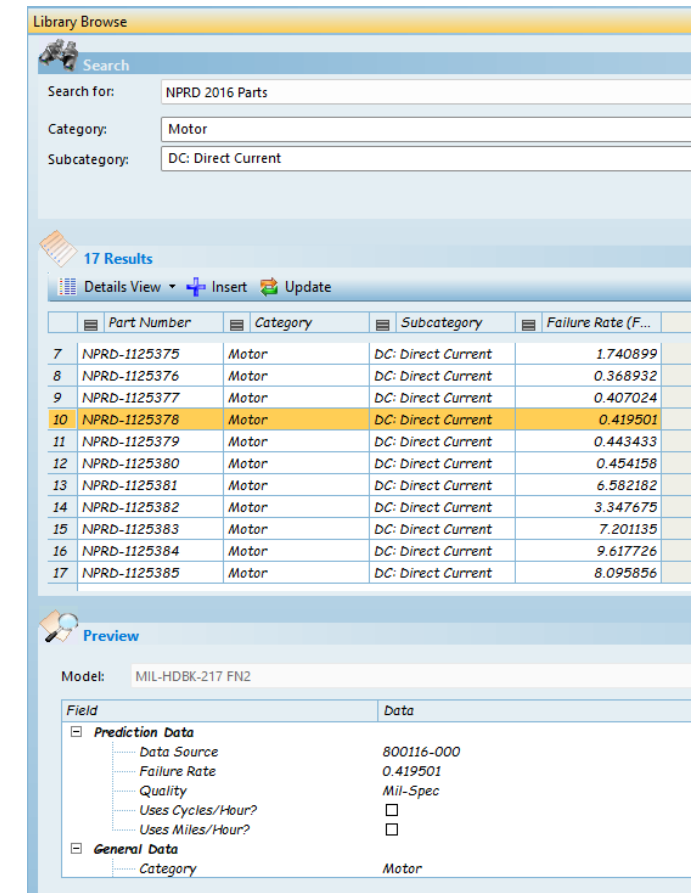


Reliability Prediction - Methods

- Reliability predictions fall into two main categories:
 - Those which make use of previous experience, design data and standard failure rate data. They include methods such as:
 - Similar Equipment – based on comparison with similar systems and conditions.
 - Generic Parts Count - based on the principle that the reliability of any item depends upon the number of parts comprising the item, the failure rates of the individual parts and the environments in which the item is to be used.
 - Parts Stress Analysis - a refinement to the Generic Parts Count method. Additionally, it requires evaluation of each part in terms of its mean operating stress levels in its intended application.
 - Those which make use of the results of testing to give practical estimates.

Reliability Prediction – Similar Equipment

- COTS reliability data can be estimated by considering other components sharing characteristics (classification, quality, nominal values...etc.) on the basis of experience and/or engineering judgement.
- Some data sources are client libraries, manufacturers and commercial libraries such as a NPRD and EPRD.
- This method is simple and quick to perform and therefore adequate in the early design stages.
- However, it must be deal rigorously since data, specially from libraries, is frequently incomplete and provided reliability figures very dispersed.



The screenshot displays the 'Library Browse' window. At the top, there is a search bar with 'NPRD 2016 Parts' entered. Below it, 'Category' is set to 'Motor' and 'Subcategory' is 'DC: Direct Current'. The results section shows '17 Results' and a table with columns: Part Number, Category, Subcategory, and Failure Rate (F...). The table lists 17 items, with item 10 (NPRD-1125378) highlighted. Below the table is a 'Preview' section for 'Model: MIL-HDBK-217 FN2', showing fields for Prediction Data (Data Source, Failure Rate, Quality, Uses Cycles/Hour?, Uses Miles/Hour?) and General Data (Category).

	Part Number	Category	Subcategory	Failure Rate (F...)
7	NPRD-1125375	Motor	DC: Direct Current	1.740899
8	NPRD-1125376	Motor	DC: Direct Current	0.368932
9	NPRD-1125377	Motor	DC: Direct Current	0.407024
10	NPRD-1125378	Motor	DC: Direct Current	0.419501
11	NPRD-1125379	Motor	DC: Direct Current	0.443433
12	NPRD-1125380	Motor	DC: Direct Current	0.454158
13	NPRD-1125381	Motor	DC: Direct Current	6.582182
14	NPRD-1125382	Motor	DC: Direct Current	3.347675
15	NPRD-1125383	Motor	DC: Direct Current	7.201135
16	NPRD-1125384	Motor	DC: Direct Current	9.617726
17	NPRD-1125385	Motor	DC: Direct Current	8.095856

Preview
Model: MIL-HDBK-217 FN2

Field	Data
Prediction Data	
Data Source	800116-000
Failure Rate	0.419501
Quality	Mil-Spec
Uses Cycles/Hour?	☐
Uses Miles/Hour?	☐
General Data	
Category	Motor

Reliability Prediction - Part Count Method

- The main benefits of the Generic Parts Count method are:
 - ✓ It allows prediction to be associated with the design process from its earliest stages and provides inputs to assist this process.
 - ✓ It is relatively quick and simple, to apply, particularly if all parts in a system are in a series reliability configuration (although pessimistic).
- Their main limitations are:
 - ✓ It assumes constant failure rate with time. Hence, the higher probabilities of failure during the early and wear-out failure periods are not considered.
 - ✓ It relies on part failure rate data derived from a variety of sources. Such data is assumed to represent average conditions, but these may vary widely.
 - ✓ Failure rate data for mechanical parts is limited, and the method does not therefore take full account of the failure contribution of such parts.

Reliability Prediction - Part Count Method

- The main steps in the Generic Parts Count method are as follows:
 1. Define the proposed system and develop a reliability model.
 2. For each block (or area) within the model:
 - a. Determine the type and number of each generic part type.
 - b. Calculate the failure rate of each generic part type for the appropriate operational conditions.
 - c. Sum the failure rates to give the total failure rate for the block (or area). Evaluate the block (or area) reliability.
 3. Combine the block (or area) reliabilities as shown by the reliability model to obtain the total system reliability



Reliability Prediction - Part Count Method

- Example of Base FR Data Table por Generic Part Count Analysis:

Component Description	Base Failure Rate λ_B failures 10 ⁶ hrs	K Factors						Predominant Failure Modes	
Part	G.1 Ground Fixed	G.2 Ground Mobile	S.1 Ship Protected	S.2 Ship Exposed	A.1 Air Protected	A.2 Air Exposed	Mode	% Factor	
Actuator, Linear (Electrical)	1.20	2.00	1.50	2.70	2.20	4.00			
Actuator, Rotary (Electrical)	2.40	2.00	1.50	2.70	2.20	4.00			
Aerial	1.00	6.00	2.30	6.00	5.00	10.00	No Transmis- sion	54	
							Signal Leakage	21	
							Spurious Transmission	25	
Aerial Dish	3.00	4.00	1.70	4.00	3.00	6.00	Ditto		
Alternator	3.00	3.00	1.30	3.00	2.00	3.00			
Batteries	3.00	3.00	1.30	3.00	2.00	3.00			
Battery, Secondary, General	1.50	10.00	8.00	10.00	24.00	50.00			
Battery, Secondary, Lead Acid	0.50	2.50	1.80	4.00	1.80	7.60	Degraded Output	70	
							Short Circuit	20	
							Intermittent Output	10	
Battery, Secondary, Nickel Cadmium	0.20	2.50	1.80	4.00	1.80	7.60	Degraded Output	72	
							No Output	28	
Battery, Secondary, Silver Zinc									
Battery, Lithium							Degraded Output	78	
							Startup Delay	14	
							Short Circuit	6	
							Open Circuit	2	
Cable, Electric, Intercon- necting - per cable (Semi- permanent Surface/ Air ground installation)	1.20	Not Applicable							
Capacitors, Fixed									
Aluminum Electrolytic	0.30	3.00	2.00	10.00	9.00	13.00	Short Circuit	50	
							Open Circuit	30	
Tantalum, Electrolytic, Foil	0.13	3.00	1.50	3.00	2.30	3.50	Short Circuit	75	
							Open Circuit	25	
Capacitors, Fixed (Continued)									

Component Description	Base Failure Rate λ_B failures 10 ⁶ hrs	K Factors					Predominant Failure Modes	
Part	G.1 Ground Fixed	G.2 Ground Mobile	S.1 Ship Protected	S.2 Ship Exposed	A.1 Air Protected	A.2 Air Exposed	Mode	% Factor
Tantalum, Electrolytic, Sintered (Anode, Wet Electrolyte)	0.12	2.40	2.30	2.40	4.80	5.00	Short Circuit	80
							Open Circuit	20
Tantalum, Electrolytic, Solid	0.08	2.30	2.00	2.30	4.00	6.00	Short Circuit	50
							Open Circuit	30
Ceramic	0.04	2.00	1.20	2.40	4.00	5.00	Short Circuit	50
							Open Circuit	30
Ceramic, Chip	0.10	2.00	1.20	1.80	1.50	2.50	Short Circuit	50
							Open Circuit	30
Mica	0.06	2.00	3.00	4.30	2.40	6.00		
Mica, Button								
Mica, Silvered								
Porcelain								
Paper, Metallized	0.06	2.00	2.70	8.00	2.00	10.00		
Paper, Foil	0.12	2.70	1.50	2.50	3.50		Short Circuit	90
							Open Circuit	5
Plastic Film (Synthetic Film)	0.05	3.00	1.70	3.00	3.00	7.00	Short Circuit	55
							Open Circuit	40
Capacitors, Variable								
Air	0.10	5.00	3.20	5.00	12.00	16.00		
Ceramic	0.14	12.00	3.00	12.00	9.00	15.00		
Glass	0.26	2.20	2.50	11.00	4.50	20.00		
Circuit Breakers								
Circuit Breaker, Magnetic	1.00	5.00	1.70	5.00	3.00	5.00		
Circuit Breaker, Thermal	2.00	2.70	1.40	2.70	2.20	7.00		
Clutch, Electro-Magnetic	3.50	3.00	2.00	3.00	4.00	5.00	Bearing Wear	30
							Coil Failure	20
							Contamination	25
							Lubrication failure	15

Reliability Prediction - Part Count Method

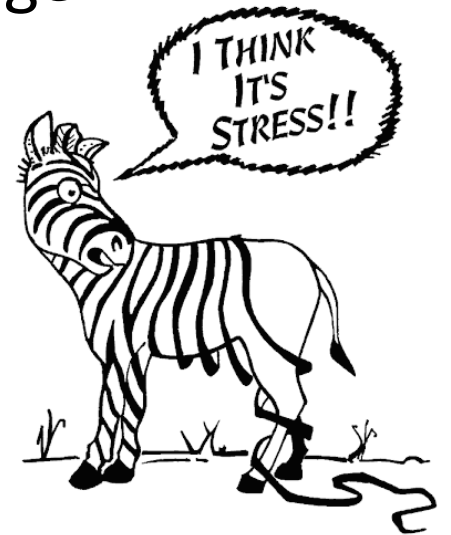
- Example of Worksheets por Generic Part Count Analysis:

(1) <u>Area of Evaluation</u>		System: SRGA Weapon System Sub-system: Control Unit Item assessed: Computer Reliability Block Diagram No: CU1			
(2) <u>Operational Conditions</u>		Mode: Operational (Search Mode) Reliability Time Interval (t): 336 hours (14 days) Environment: Ground Mobile, G2			
(3) Part Population		(4) Part Failure Rate (failures per 10 ⁶ hours)			
Parts Description	Qty N	Base Rate λ_B	Env Factor K_E	Predicted Rate $N\lambda_B K_E$	Percentage Contribution %
Capacitors, Tantalum, Solid	15	0.08	2.3	2.76	3.8
Transistors, Silicon, Signal, < 1W	50	0.05	4.4	11.00	15.2
Transistors, Silicon, High Power, > 5W	10	0.09	5.0	4.50	6.2
Diodes, Rectification, Low Power	20	0.12	2.5	6.00	8.3
Integrated Circuits, Digital, < 20 Gates	200	0.03	3.1	18.60	25.8
Resistors, Metal Oxide Film	230	0.02	4.0	18.40	25.5
Connections, Soldered, Hand	1000	0.004	1.0	4.00	5.6
Connections, Welded	2000	0.0017	1.0	3.40	4.7
Connector, Sealed (70 Pins)	1	0.025 x 70	2.0	3.50	4.9
(5) Predicted Item Failure Rate per 10 ⁶ hours = $\Sigma N\lambda_B K_E =$ 72.16					100%
(6) Item Reliability $R(t) = \exp(-\lambda t)$ $R_{CU1} = \exp(-72.16 \cdot 10^{-6} \cdot 336)$ $= \exp(-0.02425)$ $R_{CU1} = 0.976046$					
Analyst:		Date of Analysis:			

(1) <u>Area of Evaluation</u>		System: SRGA Weapon System Sub-system: Control Unit Item assessed: Computer Reliability Block Diagram No: CU1			
(2) <u>Operational Conditions</u>		Mode: Non-operational (Storage) Reliability Time Interval (t): 4368 hours (6 months) Environment: Ground Fixed Exposed Storage (GFE)			
(3) Part Population		(4) Part Failure Rate (failures per 10 ⁶ hours)			
Parts Description	Qty N	Base Rate λ_B		Predicted Rate $N\lambda_b$	Percentage Contribution %
Capacitors, Tantalum, Solid	15	0.002		0.03	3.1
Transistors, Silicon, Signal, < 1W	50	0.003		0.15	15.7
Transistors, Silicon, High Power, > 5W	10	0.005		0.05	5.3
Diodes, Rectification, Low Power	20	0.0012		0.024	2.5
Integrated Circuits, Digital, < 20 Gates	200	0.003		0.60	62.9
Resistors, Metal Oxide Film	230	0.0002		0.046	4.8
Connections, Soldered, Hand	1000	0.00002		0.02	2.1
Connections, Welded	2000	0.00001		0.02	2.1
Connector, Sealed (70 Pins)	1	0.0002 x 70		0.014	1.5
Item Base Failure Rate = $\Sigma N\lambda_b =$				0.954	100%
(5) Predicted Item Failure Rate = $(K_E \Sigma N\lambda_b = 2 \cdot 0.954)$ = 1.908 failures per 10 ⁶ hours					
(6) Item Reliability $R(t) = \exp(-\lambda t)$ $R_{CU1} = \exp(-1.908 \cdot 10^{-6} \cdot 4368336)$ $= \exp(-0.02425)$ $R_{CU1} = 0.9917$					
Analyst:		Date of Analysis:			

Reliability Prediction - Part Stress Method

- The Parts Stress Analysis method is a refinement to the Generic Parts Count method in that it involves the same basic steps. Additionally, however, it requires evaluation of each part in terms of mean operating stress levels in its intended application.
- The Parts Stress Analysis method can be used from that stage in a project when detailed design data becomes available.



Reliability Prediction - Part Stress Method

- The main steps in the Part Stress method are as follows:
 1. Define the proposed system and develop a reliability model.
 2. For each block (or area) within the model:
 - a. List each part comprising the item(s) represented by the block.
 - b. Determine the mean operating stress levels for each part and hence the factors appropriate to its particular failure rate model.
 - c. Calculate the failure rate of each part using the appropriate model.
 - d. Sum the failure rates to give the total failure rate for the block (or area). Evaluate the block (or area) reliability.
 3. Combine the block (or area) reliabilities as shown by the reliability model to obtain the total system reliability

Reliability Prediction – MIL-HDBK-217F Part Model

- Although outdated (last revision goes back to 1995), it is accepted and used worldwide.
- It supports both parts count and parts stress predictions. The first one assumes typical operating conditions of part complexity, ambient temperature, various electrical stresses, operation mode and environment. By contrast, the part stress method requires specific conditions (called Pi factors).
- It only provides the steady state failure rate in failures per operating hour.
- As mitigation for its obsolescence, ANSI/VITA 51.1 is a collaborative industry standard that provides modifications to reflect more updated and realistic failure rate assessments.

Reliability Prediction – FIDES Part Model

- A new reliability data handbook (available since January 2004) developed by a consortium of French industry under the supervision of the French DoD (DGA).
- Based on the physics of failures and supported by the analysis of test data, field returns and existing modelling.
- The FIDES 2009 model supports families count, parts count, and parts stress predictions.
- FIDES models provides average failures per calendar hour so they cannot be directly mixed with other models, such as MIL-HDBL-217F, or use their methods.

Reliability Prediction – Telcordia Part Model

- Based on various issues of the Telcordia document, Reliability Prediction Procedure for Electronic Equipment, Technical Reference SR-332.
- Last edition (issue 4) from 2016.
- Similar to MIL_HDBK-217F but using less parameters.
- Telcordia models support various calculation methods that take into account burn-in, laboratory, and/or field data.

Thank you!!!

